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RESEARCH PAPER

A hierarchical lectin-based multimodal workflow for spatial mapping of extracellular matrix glycans in fibrotic hearts

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Abstract

Cardiac fibrosis with excessive extracellular matrix (ECM) accumulation is a hallmark of hypertensive heart failure. Characterizing glycan alterations within fibrotic ECM requires analytical strategies that integrate molecular detection with spatial and structural information. However, spatial glycomics approaches applicable to ECM-rich tissue regions remain limited. Here, we established an integrated lectin-based spatial glycomics workflow combining lectin microarray–based tissue glycome mapping, lectin histochemical staining, and low-vacuum scanning electron microscopy (LVSEM). This hierarchical analytical design enables lectin signal detection, spatial localization, and structural visualization within the same tissue section. Using this workflow, we mapped *N*- and *O*-glycan distribution in fibrotic regions of hypertensive heart failure rat hearts. Among 45 lectins, *Maackia amurensis* hemagglutinin (MAH)–reactive sialo-*O*-glycans were markedly increased in WFA-positive fibrotic regions compared with controls. Glycosidase pretreatment selectively reduced WFA and MAH signals, confirming their specificities for *N*- and *O*-glycans, respectively. High-resolution single-slide imaging combining lectin fluorescence histochemistry with LVSEM revealed distinct spatial partitioning of WFA and MAH signals within fibrotic ECM regions. Co-staining with ECM glycoproteins, including periostin, collagen VI α6 chain, cartilage intermediate layer protein 1, and thrombospondin 4, showed partial spatial overlap with lectin signals, suggesting that multiple ECM glycoproteins may contribute to the observed glycan signals. These results demonstrate a lectin-based spatial glycomics workflow integrating glycomic signal detection, spatial localization, and ultrastructural validation, establishing a lectin-based multimodal analytical framework for spatially resolved glycan partitioning in fibrotic tissues.

Keywords Cardiac fibrosis · Spatial glycomics · Tissue glycome mapping · *z*-Axis imaging

Introduction

Glycosylation serves multiple biological functions, and alterations in the glycosylation machinery are associated with many diseases, including cardiac diseases [1]. However, although aberrant glycosylation has been implicated in fibrotic extracellular matrix (ECM) tissues, how glycosylation is spatially remodeled within ECM-rich fibrotic regions remains poorly understood, largely due to the lack of analytical platforms that can simultaneously resolve glycan heterogeneity and tissue architecture. Fibrotic tissues are dominated by dense ECM structures in which parenchymal cells are partially or completely replaced [2]. This makes conventional nucleic acid–based spatial omics approaches insufficient to directly resolve glycan structures within ECM-rich fibrotic regions. Moreover, glycosylation is a complex post-translational modification regulated by multiple enzymatic

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processes [3]; therefore, glycan structures cannot be directly predicted from transcriptomic data. Thus, analytical strategies capable of integrating glycomic profiling with spatial tissue information are essential for understanding glycan remodeling during fibrosis.

Cardiac fibrosis is a typical pathological feature of heart failure and other cardiovascular diseases [2]. It is characterized by excessive deposition of ECM components, which alters tissue architecture and contributes to organ dysfunction. Because fibrotic regions are dominated by ECM-rich structures in which cardiomyocytes are partially or completely replaced, understanding molecular alterations in the ECM is essential for elucidating fibrosis-related pathophysiology [2, 4, 5]. Several studies have reported disease-associated glycosylation changes in cardiac pathology. For instance, mucin-type *O*-glycosylation and core defucosylation of *N*-glycans have been reported to be significantly altered in a rat model of hypertension-induced cardiac hypertrophy and heart failure [6]. In addition, ECM proteins such as periostin (POSTN) and collagen VI α6 chain (COL6A6) have been identified as glycoproteins carrying fibrosis-associated glycans recognized by *Wisteria floribunda* agglutinin (WFA) in a dilated cardiomyopathy model (DCM) [7]. These findings suggest that ECM proteins in fibrotic tissues may carry disease-specific glycan signatures.

Lectin microarray (LMA) has been widely employed due to its high sensitivity in glycomic profiling of glycoproteins, enabling simultaneous analysis of both *N*- and *O*-glycans [8–10]. When combined with laser microdissection (LMD), LMA enables performing region-specific glycomic profiling in formalin-fixed paraffin-embedded (FFPE) tissues. The LMD-assisted LMA procedure has been standardized for spatial glycomic analysis [11], including at the level of specific cell types [12]. This technique has been applied to the study of cardiac fibrosis in a DCM mouse model, leading to the identification of WFA as a lectin useful for detecting fibrosis-associated *N*-glycosylation as previously reported [7]. However, although lectin microarray–based glycome mapping can identify glycan signatures associated with defined tissue regions, it cannot directly visualize glycan localization within the ultrastructural architecture of fibrotic ECM.

To address this challenge, we established a hierarchical lectin-based multimodal spatial glycomics workflow integrating glycomic detection, spatial localization, and ultrastructural validation. We applied this workflow to a hypertensive heart failure rat model, whose pathophysiology and RNA-seq profiles have been previously reported [13]. We first confirmed that WFA was enriched in fibrotic regions in this model and could be used to guide region selection for spatial glycomics analysis. We then performed tissue glycome mapping with WFA-guided fibrosis region-specific sampling to compare glycomic profiles between high-salt (HS)-fed and low-salt (LS)-fed control rats. For spatial glycomic analysis, we employed LMD-assisted LMA with improved sample preparation and data acquisition using a highly sensitive scanner [14]. This approach enabled the detection of lectin signals associated with fibrotic regions, including *Maackia amurensis* hemagglutinin (MAH)–reactive sialo-*O*-glycans and WFA-reactive asialo-*N*-glycans. To further examine the spatial distribution of these glycans, we employed low-vacuum scanning electron microscopy (LVSEM) [15], which enabled visualization of glycan localization within fibrotic tissue architecture. Finally, based on previous reports of DCM [7], RNA-seq datasets of this model [13], and database analyses (see "Materials and methods"), we examined ECM glycoproteins showing partial spatial overlap with WFA and MAH signals in fibrotic regions. Together, this study presents a hierarchical lectin-based spatial glycomics workflow integrating tissue glycome mapping, lectin histochemistry, and LVSEM imaging, providing an analytical framework for spatially mapping of ECM glycans in fibrotic tissues.

Materials and methods

Animal procedures

All tissue samples were obtained from Dahl-Iwai S rats subjected to HS and LS diets, serving as hypertensive heart failure and control models, respectively, which were prepared in a previous study [13]. Briefly, the LS and HS rats were fed 0.3% and 0.8% NaCl diets, respectively, after 6 weeks of age. Untreated 18-week-old HS and age-matched LS rats were used for heart tissue collection. At 18 weeks of age, HS rats ($N=7$) exhibited severe hypertension (systolic blood pressure of 210.5 ± 9.0 and 152.0 ± 3.2 mmHg for HS and LS rats, respectively) and heart failure (left ventricular [LV] ejection fraction of 46.2 ± 2.6 and 61.1 ± 1.1% for HS and LS rats, respectively) compared to LS rats ($N=6$). Data are presented as mean ± SEM.

Tissue glycome mapping

This study employed a lectin-based tissue glycome mapping workflow using LMD-assisted LMA. This methodology was originally developed in our previous study [11], and a detailed experimental protocol was subsequently described [16]. An improved platform incorporating a highly sensitive scanner was later reported [14], and an updated protocol for this improved workflow has recently been published [17]. The procedures applied in the present study were performed according to these established and updated protocols. The detailed experimental procedures are described below.

Tissue section preparation for tissue glycome mapping

The heart tissues were fixed in 10% neutral-buffered formalin and embedded in paraffin, as previously described [13]. For tissue glycome mapping, tissue sections were prepared according to established protocols [11, 16, 17]. Briefly, 5-µm-thick sections on polyethylene naphthalate (PEN) slides were deparaffinized and rehydrated using an ethanol gradient. The sections were washed with water, stained with Mayer's hematoxylin solution (Wako, Osaka, Japan), rinsed under running water, air-dried at room temperature (RT), and subjected to LMD within 1 week.

Tissue dissection and protein extraction

Tissue dissection was performed using an LMD system (LMD7000; Leica Microsystems, Wetzlar, Germany) coupled to a DM6000B microscope (Leica Microsystems). The procedure was conducted as previously described [14, 17]. Briefly, tissue fragments (0.5 mm^2 total area) were collected from hematoxylin-stained sections into 0.5-mL protein low-binding tubes (BT-060L, INA OPTIKA, Osaka, Japan). The fragments were centrifuged at 18,000 × g for 1 min at 4 °C, followed by antigen retrieval in 200 µL of 10 mM citrate buffer (pH 6.0) for 1 h at 95 °C. Subsequently, 190 µL of supernatant was discarded and the fragments were washed with 190 µL of D-PBS. After removing all the buffer, the fragments were subjected to protein extraction using 10 µL of D-PBS containing 0.5% Nonidet P-40 (NP-40). The extraction process included sonication (three times, 3 s each), incubation on ice for 1 h, and centrifugation at 18,000 × g for 1 min at 4 °C to collect the supernatants as protein extracts.

LMA analysis

Due to the limited material obtained by LMD, direct quantification of total protein was not feasible. Therefore, equal tissue area (0.5 mm^2 per region) was used as a standardized input substitute to minimize variation in sample loading [11, 14]. LMA of heart tissue fragments was performed as follows: 10 µL protein extract obtained from a 0.5 mm^2 tissue fragment was fluorescently labeled with Cy3-mono-reactive dye (Cytiva, Marlborough, MA) for 1 h at RT in the dark. The labeling reagent was supplied as "sufficient for 1 mg protein per vial." An amount corresponding to the labeling capacity for approximately 10 µg of protein was used according to the manufacturer's instructions. After adjusting the volume to 100 µL with probing buffer (25 mM Tris-HCl at pH 7.5, 137 mM NaCl, 2.7 mM KCl, 500 mM glycine, 1 mM $CaCl_2$, 1 mM $MnCl_2$, and 1% Triton X-100), the mixtures were incubated for 2 h at RT to quench the excess reagent. Then, 60 µL of labeled samples (0.3 mm^2 equivalent) was loaded onto the lectin array chip (LecChip Ver.1.0; GlycoTechnica, Yokohama, Japan) containing triplicate spots of 45 lectins (the abbreviations and carbohydrate specificities of the lectins on the array are summarized in Supplementary Table 1) and incubated at 20 °C with 60 rpm shaking overnight. After washing thrice with probing buffer, the LecChip was scanned using an evanescent-field fluorescence scanner (GlycoStation Reader 2300; Emukk, Yokohama, Japan). All data were analyzed using the GlycoStation Tools Pro Suite 2.0 software (GlycoTechnica). The data acquired under appropriate exposure times with net intensities < 40,000 for all lectin spots were used to obtain the glycomic profiles as the relative signal intensities of the 45 lectins normalized to the mean value of all lectin signals [11, 14].

Histological staining

Collagen staining

Collagen staining using Masson's trichrome (MT) and Sirius Red (SR) was performed as previously described [13].

Single lectin staining

Lectin staining was performed as described previously [7]. Heart FFPE (3 µm) tissue sections were used for antigen retrieval (pH 6; Dako, Carpinteria, CA); then, the sections were blocked with the streptavidin-biotin blocking kit (Vector Laboratories, Burlingame, CA) and further blocked with a carbo-free blocking solution (Vector Laboratories) for 1 h at RT. The sections were incubated overnight at 4 °C with a biotinylated lectin (WFA, MAH, and ACA; Vector Laboratories) at a final concentration of 2 µg/mL in PBS. After washing with PBS, the sections were further incubated with streptavidin-conjugated Alexa Fluor 647 (1 µg/mL in PBS; Thermo Fisher Scientific, Waltham, MA), a membrane marker wheat germ agglutinin (WGA; fluorescein isothiocyanate [FITC]–conjugated, 2 µg/mL in PBS; Vector Laboratories), and Hoechst 33342 (0.5 µg/mL in PBS; Dojindo, Kumamoto, Japan), for 1 h at RT. The autofluorescence in the sections was then quenched using a TrueVIEW autofluorescence quenching kit (Vector Laboratories), mounted using ProLong Diamond antifade reagent (Thermo Fisher Scientific), and scanned using a digital slide scanner (NanoZoomer S60; Hamamatsu Photonics, Hamamatsu, Japan) or a fluorescence microscope (BZ-X710; Keyence, Osaka, Japan).

Fibrotic area quantification

The area quantification was performed using measurement software (Hybrid Cell Count and Macro Cell Count;

Keyence). The ratio of collagen-positive (SR^+ and MT^+) area to the total SR- and MT-stained area was calculated. Similarly, the WFA^+ and MAH^+ areas were determined by calculating the ratio of their respective stained areas, with adjustments made by subtracting the WGA-stained area (approximating the tissue area) of the section.

$$WFA\ quantitative\ area = \left(\frac{[WFA\ positive\ area\ (\mu m^2)]}{[WGA\ positive\ area\ (\mu m^2)]} \right) \times 100\%$$

$$MAH\ quantitative\ area = \left(\frac{[MAH\ positive\ area\ (\mu m^2)]}{[WGA\ positive\ area\ (\mu m^2)]} \right) \times 100\%$$

Double lectin staining

Double staining of WFA and MAH lectins was performed as follows: After antigen retrieval and blocking, the sections were incubated overnight at 4 °C with a first biotinylated lectin (2 µg/mL in PBS; Vector Laboratories). Then, they were washed three times with PBS and incubated with a second lectin (2 µg/mL in PBS) that was manually labeled with HiLyte Fluor 647 using a labeling kit (Dojindo, Kumamoto, Japan) at 20 °C for 6 h. Then sections were washed with PBS, and further incubated with streptavidin-conjugated Alexa Fluor 555 or Cy3 (Thermo Fisher Scientific; 1 µg/mL in PBS) and the membrane marker WGA (CF350-conjugated; 1 µg/mL in PBS; Biotium, Hayward, CA), for 1 h at RT. The stained sections were mounted using the ProLong Diamond Antifade reagent.

Selection of candidate glycoproteins for spatial co-distribution analysis

Based on our previous report [7], enrichment in WFA-bound fractions of DCM mouse hearts, periostin (POSTN) and collagen VI α6 chain (COL6A6) were selected as candidate ECM glycoproteins for co-staining with WFA. These secreted ECM glycoproteins, localized in extracellular regions where WFA positivity is typically observed, were confirmed by UniProt (https://www.uniprot.org/) and Human Protein Atlas (https://www.proteinatlas.org/) databases [18, 19].

For double staining with MAH, the candidate extracellular or secreted proteins were sorted from RNA-seq data [13]. The selection was based on the following criteria: (i) differential expression in HS versus LS ($\log_2$ fold change > 2 and $P < 0.05$), and (ii) predicted extracellular or secreted localization with possible *O*-GalNAcylation sites according to GlyGen (https://glygen.org) [20], UniProt [18], and Human Protein Atlas [19] databases.

Double staining using antibodies and lectins

The staining using antibodies and lectins was performed as follows: After antigen retrieval at pH 6 for POSTN, COL6A6, and N-terminal pro-atrial natriuretic peptide (NT-proANP) or pH 9 for thrombospondin 4 (THBS4) and cartilage intermediate layer protein 1 (CILP) and blocking, the sections were incubated overnight at 4 °C with a biotinylated lectin (WFA or MAH, 2 µg/mL in PBS; Vector Laboratories). Next, the sections were washed with PBS, and incubated with antibodies against COL6A6 (10 µg/mL in PBS; MyBioSource, CA), POSTN (5 µg/mL; Novus Biologicals, CO), NT-proANP (1 µg/mL in PBS; #592-12), THBS4 (4 µg/mL; Affinity Biosciences, OH), and CILP–Alexa Fluor 555 (4 µg/mL; Bioss, Boston, MA) at 20 °C for 6 h. The sections were washed three times with PBS, followed by incubation for 1 h at RT with streptavidin-conjugated Alexa Fluor 647 (1 µg/mL in PBS); a secondary antibody (Alexa Fluor 555-conjugated anti-rabbit IgG, 2 µg/mL in PBS; Thermo Fisher Scientific); the membrane marker WGA (FITC-conjugated; 2 µg/mL in PBS); and Hoechst 33342 (0.5 µg/mL in PBS). The stained sections were mounted using the ProLong Diamond Antifade reagent.

Glycosidase treatment

After deparaffinization and antigen retrieval, *N*-glycans on the serial tissue sections were digested using peptide-*N*-glycosidase F, which was diluted with 1× reaction buffer (50 mM sodium phosphate, pH 7.5) to a final concentration of 500 units (total/slide) (catalog number: NZS1, N-Zyme Scientifics, Doylestown, PA), and incubated at 37 °C for 16 h. For *O*-glycan removal, the tissue sections were treated with α2–3,6,8 sialidase 500 units (total/slide) (catalog number: GK80040, New England BioLabs) and *O*-glycosidase 4000 units (total/slide) (catalog number: P0733, New England BioLabs) in 1× reaction buffer (50 mM sodium phosphate, pH 7.5) containing 0.5% NP-40 at 37 °C for 16 h, as described elsewhere [21]. After washing thrice with PBS, the sections were processed to double staining using WFA and MAH, as described above.

LVSEM imaging and histochemical staining

Single-slide imaging of lectin fluorescence and LVSEM

Single-slide imaging was performed as previously described [15], with minor modifications. After fluorescence image acquisition, the same tissue sections were subjected to histochemical staining and subsequent low-vacuum scanning electron microscopy (LVSEM), enabling spatial correlation of lectin fluorescence and ultrastructural topology on a single slide. This integrated workflow was designed to evaluate

the analytical feasibility of combining lectin-guided fluorescence microscopy with LVSEM-based ultrastructural imaging for spatial glycomic partitioning within localized tissue regions.

Histochemical staining protocols for LVSEM

Three staining protocols were evaluated for compatibility with LVSEM imaging: Masson's phosphotungstic acid (Masson-PTA), reticulin staining, and a modified periodic acid–methenamine silver (PAM) protocol optimized for LVSEM-based contrast enhancement [15].

Masson-PTA staining

After fluorescence imaging, sections were stained for 10 min with Masson's solution (ponceau xylidine, acid fuchsin, azophloxine; Muto Pure Chemicals, Tokyo, Japan). After rinsing in distilled water, sections were immersed in 2.5% phosphotungstic acid for 60 min at RT. Sections were subsequently washed, dehydrated through graded ethanol, and air-dried.

Reticulin staining

Sections were oxidized with potassium permanganate ($KMnO_4$) according to standard reticulin protocols, followed by immersion in ammoniacal silver solution under controlled temperature conditions. After reduction and optional gold toning, sections were fixed with sodium thiosulfate, washed thoroughly, dehydrated, and air-dried.

Modified PAM staining

The modified PAM protocol was performed as previously described [15], based on Gomori–Jones methenamine silver staining with LVSEM-oriented modifications. In brief, paraffin sections (3 µm thickness) were oxidized with 0.5% periodic acid for 10 min, followed by treatment with 0.5% thiosemicarbazide for 5 min to enhance silver binding. Sections were then incubated in freshly prepared methenamine-silver solution at 65 °C for 30–40 min. After confirmation of adequate basement membrane darkening under light microscopy, sections were subjected to overnight gold chloride treatment to stabilize silver deposits and enhance electron density. Samples were then fixed in photographic fixing solution, thoroughly washed, dehydrated, and air-dried. Detailed reagent preparation and optimization parameters are provided in our previous methodological report [15].

Principle of contrast enhancement

Lectin fluorescence labeling was rendered detectable under LVSEM through heavy metal sensitization involving silver enhancement and subsequent gold substitution. Deposition of high atomic number metals increases backscattered electron yield, thereby enabling ultrastructural visualization of lectin-reactive regions under low-vacuum conditions without conductive coating. This approach allows direct spatial correlation between fluorescence signals and extracellular matrix architecture in three dimensions. Accordingly, LVSEM imaging in this study was used for ultrastructural validation of lectin signals rather than for quantitative assessment of glycan abundance.

Statistical analysis

All statistical analyses, including principal component analysis (PCA), were performed using JMP software (Student Edition version 18; SAS Institute, Cary, NC, USA). The net intensities obtained by LMA under the same exposure time for HS and LS rat heart samples were assessed for normality using the Shapiro–Wilk test. Non-normally distributed data were analyzed using nonparametric methods. Normalized glycomic profiles were subjected to PCA as an exploratory tool to visualize the overall variance structure. Sample inclusion or exclusion was determined a priori based on predefined technical criteria recorded in the experimental metadata (e.g., independent processing days), rather than on PCA coordinates. Statistical comparisons of glycomic profiles were conducted using the Mann–Whitney *U*-test (for two groups), and the Steel–Dwass all-pairs test for the LS–WFA^-, HS–WFA^-, and HS–WFA^+ groups. A two-sided $P < 0.05$ was considered statistically significant. The correlation of collagen deposition (MT positivity) and lectin-positive areas for WFA and MAH were evaluated using the Spearman rank correlation test.

Results and discussion

To implement this hierarchical lectin-based spatial glycomics workflow (as summarized in the Graphical abstract), we sequentially applied three analytical steps corresponding to glycomic signal detection (Workflow step [WS]1), spatial localization (WS2), and ultrastructural validation (WS3). In WS1, we performed WFA-guided tissue glycome mapping using LMD-assisted LMA to detect fibrosis-associated glycan signals in defined regions. In WS2, we verified the spatial localization and biochemical specificity of the detected lectin signals by lectin histochemistry combined with glycosidase pretreatment. In WS3, we finally examined the distribution of these lectin

signals within the ultrastructural architecture of the ECM using single-slide LVSEM imaging.

WS1: Glycomic signal detection by tissue glycome mapping of fibrotic and non-fibrotic regions in hypertensive heart failure

WFA has been reported to be a +useful lectin for detecting cardiac fibrosis in a DCM mouse model [7]. To determine whether WFA-guided detection is also applicable to hypertensive heart failure, coronal heart sections from 18-week-old HS and LS rats were stained with WFA. To compare relative glycomic profile changes between microdissected fibrotic and non-fibrotic areas, serial sections of HS and LS hearts were stained with WFA to define regions for LMD-assisted tissue glycome mapping. In HS rat hearts, WFA staining is positive within fibrotic areas defined by collagen staining (MT and Sirius Red) in the LV walls, whereas WFA staining is almost undetectable in LS rat hearts (Fig. 1a and Supplementary Fig. 1a). This observation supports the use of WFA reactivity to guide fibrotic region selection for spatial glycomic analysis. Notably, the fibrotic areas in HS rat hearts were smaller and more patchily distributed than those in the DCM model [7]. However, leveraging the high sensitivity of LMD-assisted LMA analysis, 0.5 mm^2 tissue fragments are sufficient for glycome mapping (Fig. 1b). Glycomic profiles are obtained from WFA^+ and WFA^- regions of HS and LS rat hearts (Supplementary Fig. 1a).

Because the aim of this workflow step is region-resolved tissue glycome mapping, the input amount was standardized based on the microdissected tissue area rather than total protein quantity, which cannot be reliably quantified from the small LMD-derived samples. The net intensities of the 45 lectins in each sample (Supplementary Table 2) were normalized to generate the glycomic profiles of each tissue region (Supplementary Table 3). For quality control within this workflow, samples showing extreme global increases or decreases in net signal intensity or batch-dependent variation were excluded from statistical analysis. This workflow has been standardized [11], and stable relative glycomic profiles have been reported across a range of input (area) amounts using the improved LMD-LMA platform in our previous study [14]. To evaluate the overall data structure and detect potential technical variation, glycomic profiles were assessed using PCA. In PCA, two LS samples (LS2 and LS3, Supplementary Fig. 2a) are clearly separated from the main sample distribution, which correspond to samples prepared in a different experimental batch. These samples were flagged as potentially batch-influenced samples and were handled through sensitivity analyses rather than exclusion. Consequently, glycomic profiles from 10 LS–WFA^-, 21 HS–WFA^-, and 17 HS–WFA^+ regions were used in subsequent analyses. These datasets therefore provided a robust basis for quantitative comparison of regional glycomic alterations associated with cardiac fibrosis.

Consistent with the PCA structure, the glycomic profiles show that HS–WFA^+ samples are separated along PC1, whereas LS–WFA^- and HS–WFA^- show substantial overlap (Fig. 1c). The glycomic profiles are then statistically compared between HS and LS rats as well as among LS–WFA^-, HS–WFA^-, and HS–WFA^+ regions (Supplementary Table 4). Among the 45 lectins (Supplementary Table 1), WFA signals do not show significant differences among the LS–WFA^-, HS–WFA^-, and HS–WFA^+ regions, as well as between the HS and LS rats, likely because of the limited fibrotic areas in the HS rat hearts. Other lectins that recognize asialo-*N*-glycans (RCA120 and DSA) also showed statistically significant differences, albeit the $\log_2$ fold change was < 1.0. In contrast, *O*-glycan-binding lectins, including MAH, HPA, MPA, and ACA, as well as TxLC-I (which binds to the terminal GalNAc), show significantly higher signals ($\log_2$ fold change; 5.9, 4.1, 1.3, 1.1, and 1.6, respectively) in the HS rats than in the LS rats (Supplementary Table 4). Since TxLC-I shares glycan epitopes with WFA, the increased TxLC-I signal is consistent with the increase in asialo-*N*-glycans observed in hypertensive heart failure. Among *O*-glycan-binding lectins, MAH shows the most significant differences ($P < 0.05$) with $\log_2$ fold changes of 5.9 (HS vs. LS), 4.9 (HS–WFA^- vs. LS–WFA^-), and 6.6 (HS–WFA^+ vs. LS–WFA^-) (Fig. 1c and Supplementary Table 4). A similar tendency is observed when LS2 and LS3 samples are included (Supplementary Fig. 2b), indicating that their exclusion does not affect the overall conclusion. This finding suggests a marked increase in sialo-*O*-glycans in the WFA^+ fibrotic areas. Collectively, WFA-reactive asialo *N*-glycan features, and MAH-reactive sialo-*O*-glycans were both increased in fibrotic cardiac tissues of this hypertensive heart failure rat model. These results define MAH and WFA as orthogonal lectin readouts for subsequent workflow steps, enabling localization- and structure-level validation of the array-derived signals. Accordingly, WFA and MAH were selected for further validation of cardiac fibrosis by histochemical staining.

MAH primarily recognizes disialylated core 1 *O*-glycans (disialyl-T), whereas ACA binds broadly to core 1 and core 2 structures, and its affinity can be enhanced by α2–3 sialylation on Gal and α2–6 sialylation on GalNAc residue (disialyl-T) [22]. In contrast, HPA and MPA preferentially bind to non-sialylated *O*-glycans and do not efficiently recognize the sialylated forms. Considering these differences in glycan specificity, the present results indicate that sialo-*O*-glycans are selectively enriched in the fibrotic regions. The increase in *O*-glycans is consistent with previous findings in failing hearts from Dahl rats [6], which demonstrated *O*-glycosylation-related gene upregulation. In a previous

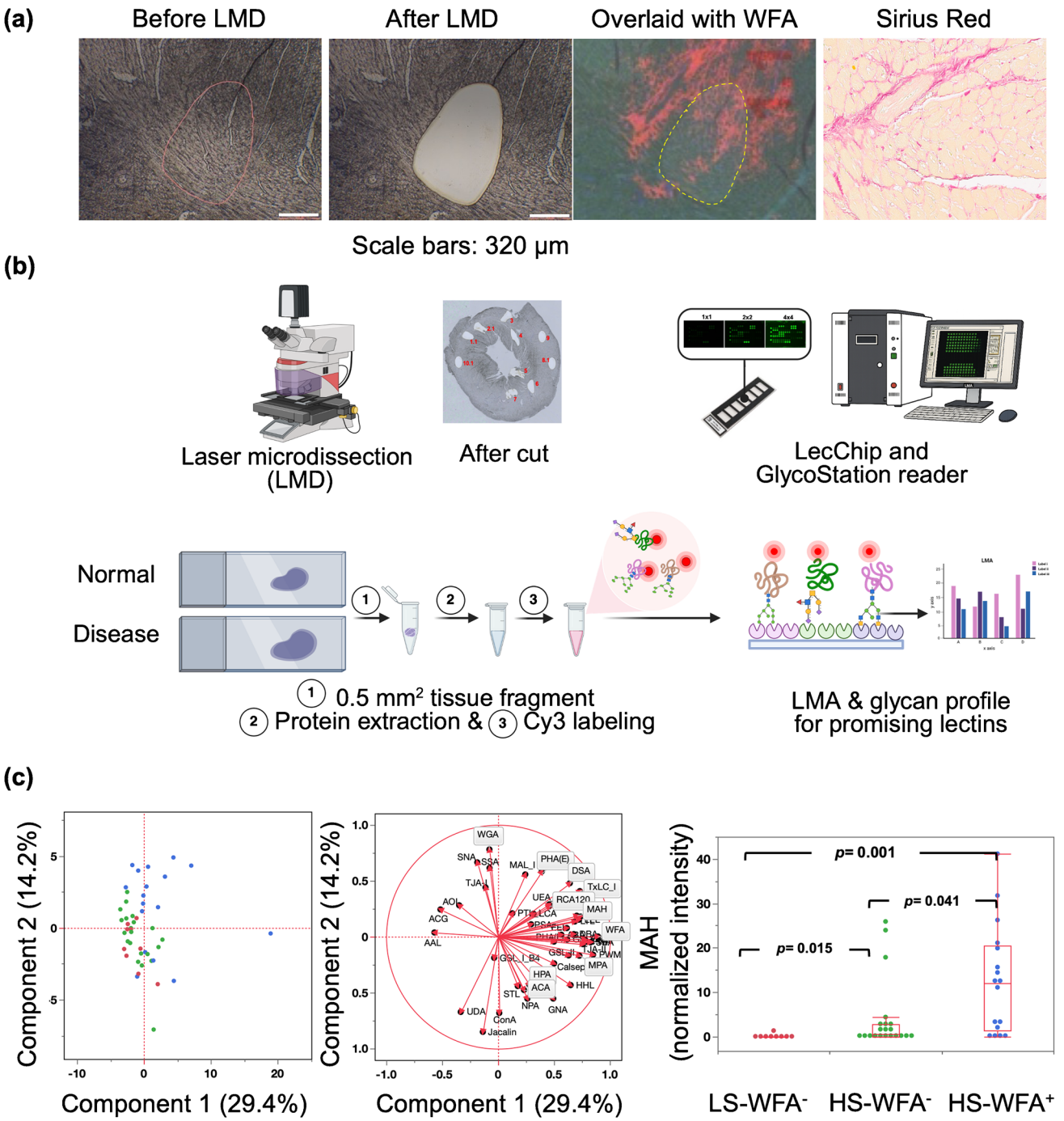


Fig. 1 Laser microdissection (LMD)–assisted lectin microarray (LMA) analysis. **a** Representative enlarged images of the targeted WFA^+ area before and after LMD (in area 1.1), along with the LMD image overlaid with WFA staining. For comparison of spatial distribution, an SR-stained section from the same individual is also shown, highlighting the corresponding region. **b** Schematic workflow of tissue glycome-mapping technique. Hematoxylin-stained formalin-fixed paraffin-embedded (FFPE) cardiac tissue sections ($n = 3$ individuals/group) from high-salt-treated (HS) and low-salt-treated (LS) rats were subjected to tissue collection using LMD ($n = 10$ fragments/individual). After protein extraction and Cy3 labeling, the resulting tissue samples were analyzed by LMA to obtain glycomic profiles. **c** Principal component analysis (PCA) and statistical analysis of the normalized intensity for *Maackia amurensis* hemagglutinin (MAH) signals. The *P* values of the Steel–Dwass test are indicated

study, LMA analysis employed lysates of frozen LV tissues that had not been defatted, possibly allowing glycolipid contamination, which may have masked localized *O*-glycan signals in limited fibrotic areas. A key feature of this study was tissue glycome mapping combined with WFA staining-based fibrotic enrichment selection, enabling comprehensive investigation of glycosylation changes in hypertensive heart failure. Based on precise sample collection and highly sensitive analysis, we identified MAH-binding *O*-glycans as an additional fibrosis-related feature. This approach has an advantage in analyzing fibrotic regions, especially large scars (replacement fibrosis), where cardiomyocytes are replaced by ECM—a feature not identified by nucleic acid–based approaches such as RNA-seq. Accordingly, these results highlight the capability of tissue glycome mapping to enable region-resolved analysis of glycosylation alterations associated with cardiac fibrosis.

WS2: Spatial localization by lectin histochemistry to verify fibrosis-associated glycan signals

To verify the array-based finding of increased MAH-reactive sialo-*O*-glycans associated with cardiac fibrosis, serial sections of HS and LS rat heart tissues were stained with MAH and WFA. WFA staining was associated with reactive fibrosis, including both interstitial and perivascular fibrotic regions adjacent to blood vessels (Supplementary Fig. 1 and Fig. 2a, b), and showed a distribution pattern distinct from that observed in the DCM mouse model [7]. In contrast, MAH staining was broadly distributed throughout the tissue sections, with particularly strong signals in regions corresponding to fibrotic tissues as confirmed by MT staining in serial sections (Fig. 2a, c). When the sections were co-stained with WFA and MAH under control conditions, their signals exhibited different patterns with only partial overlap (appearing as white in Fig. 3; WFA/MAH, Control) in the areas of most severe fibrosis. Consistent with the LMA results, the MAH staining pattern suggests that MAH-reactive sialo-*O*-glycans features are enriched in fibrotic regions.

To assess the spatial relationship between collagen deposition and aberrant glycosylation, we quantitatively compared the areas positive for classical collagen stains (SR^+ and MT^+) with those positive for WFA and MAH throughout the LV coronal section in the same individuals analyzed by LMD-assisted LMA. Similar to the previous report using MT-stained sections [13], significant differences are observed in SR^+ fibrotic areas between the LS and HS rat tissues (Supplementary Fig. 3a). MAH staining exhibited a broad distribution pattern with dense signals in fibrotic areas, whereas WFA staining (Supplementary Fig. 1b) is more specifically localized to fibrotic regions. The MAH-positive area tended to increase with fibrotic area, whereas the WFA-positive area showed a weaker trend. Spearman's rank correlation analysis showed a positive correlation for MAH ($\rho = 0.841$, $P = 0.036$) and a weaker, non-significant correlation for WFA ($\rho = 0.626$, $P = 0.183$). Notably, MAH^+ and WFA^+ regions were smallest in the HS3 sample, indicating that this HS individual exhibited less fibrosis than the LS controls. Taken together, these observations further support that the glycosylation recognized by these lectins is increased in fibrotic tissues.

Investigation of glycan epitopes attributable to fibrosis-selective lectin staining

Based on the previous findings in a DCM mouse model [7], WFA staining was considered to result from binding to asialo-*N*-glycans. To validate this assumption, heart tissue sections from HS and LS rats were treated with PNGase F to eliminate *N*-glycans and then stained with WFA and MAH for comparison. After PNGase F treatment, WFA staining is markedly reduced, whereas MAH staining is largely preserved (Fig. 3a). In contrast, after sialidase and *O*-glycosidase treatment, MAH staining is nearly abolished (Fig. 3b). These results support that MAH staining results from binding to sialo-*O*-glycans. Conversely, WFA staining was increased after sialidase and *O*-glycosidase treatment, demonstrating that the glycans responsible for WFA binding were not *O*-glycans. Collectively, these results indicate that asialo-*N*-glycans and sialo-*O*-glycans are responsible for WFA and MAH staining, respectively, in fibrotic tissues, thus providing biochemical support for the fibrosis-associated lectin reactivity.

WS3: Ultrastructural validation by single-slide lectin fluorescence histochemistry combined with LVSEM

Given the biochemical specificity of WFA and MAH, we next elucidated whether these glycans exhibit distinct or overlapping spatial localization within fibrotic lesions. Although classical fluorescence microscopy revealed limited partial overlap of WFA and MAH staining within fibrotic areas, this apparent overlap may result from the two-dimensional projection of spatially segregated glycans. Therefore, we examined their ultrastructural spatial distribution using high-resolution LVSEM on the same slides used for the double fluorescence staining. A key feature of this workflow is that the same slide used for lectin fluorescence histochemistry can be directly subjected to LVSEM, enabling structure-level interpretation of lectin signals within the same tissue context.

To assess the interstitial architecture of these regions, we first compared three histochemical staining methods: Masson's PTA, a variant of the widely used MT; reticulin staining; and the modified PAM method [15], which was

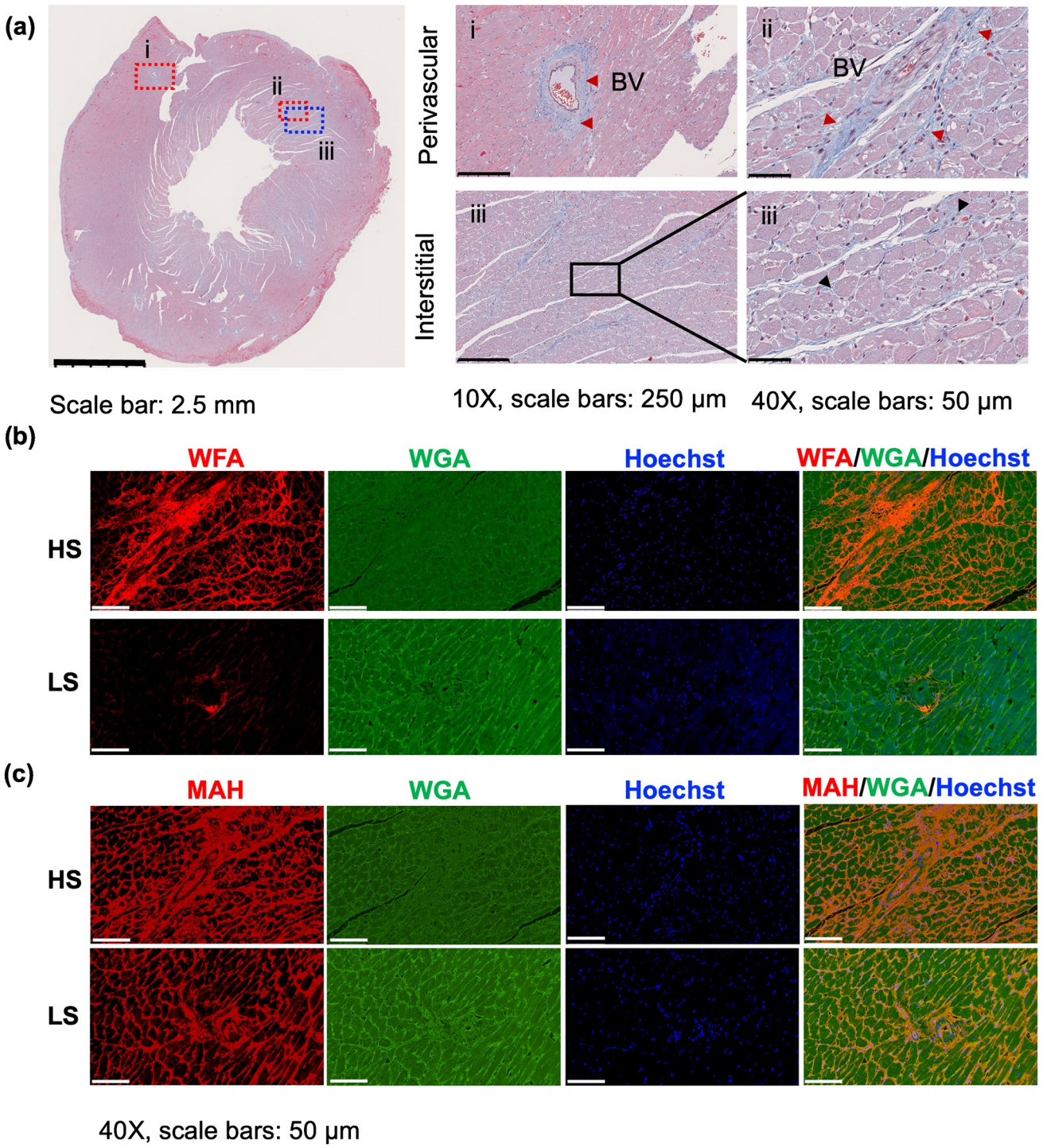


Fig. 2 Lectin histochemistry of *Wisteria floribunda* agglutinin (WFA) and *Maackia amurensis* hemagglutinin (MAH) in rat hearts. **a** Representative Masson's trichrome (MT)–stained section of an HS rat heart. Fibrotic regions adjacent to a blood vessel (BV) and interstitial fibrotic regions are indicated (arrowheads) and shown at higher magnification in panels i–iii. **b** Representative images of WFA staining in fibrotic regions adjacent to a blood vessel in serial sections of HS and LS rat hearts (corresponding to the MT region in a-ii), shown together with wheat germ agglutinin (WGA) and Hoechst, as well as the merged image. **c** Representative images of MAH staining in the corresponding region of HS and LS rat hearts, shown together with WGA and Hoechst, as well as the merged image. WGA and Hoechst were used to visualize membranes and nuclei, respectively. Scale bars are as indicated

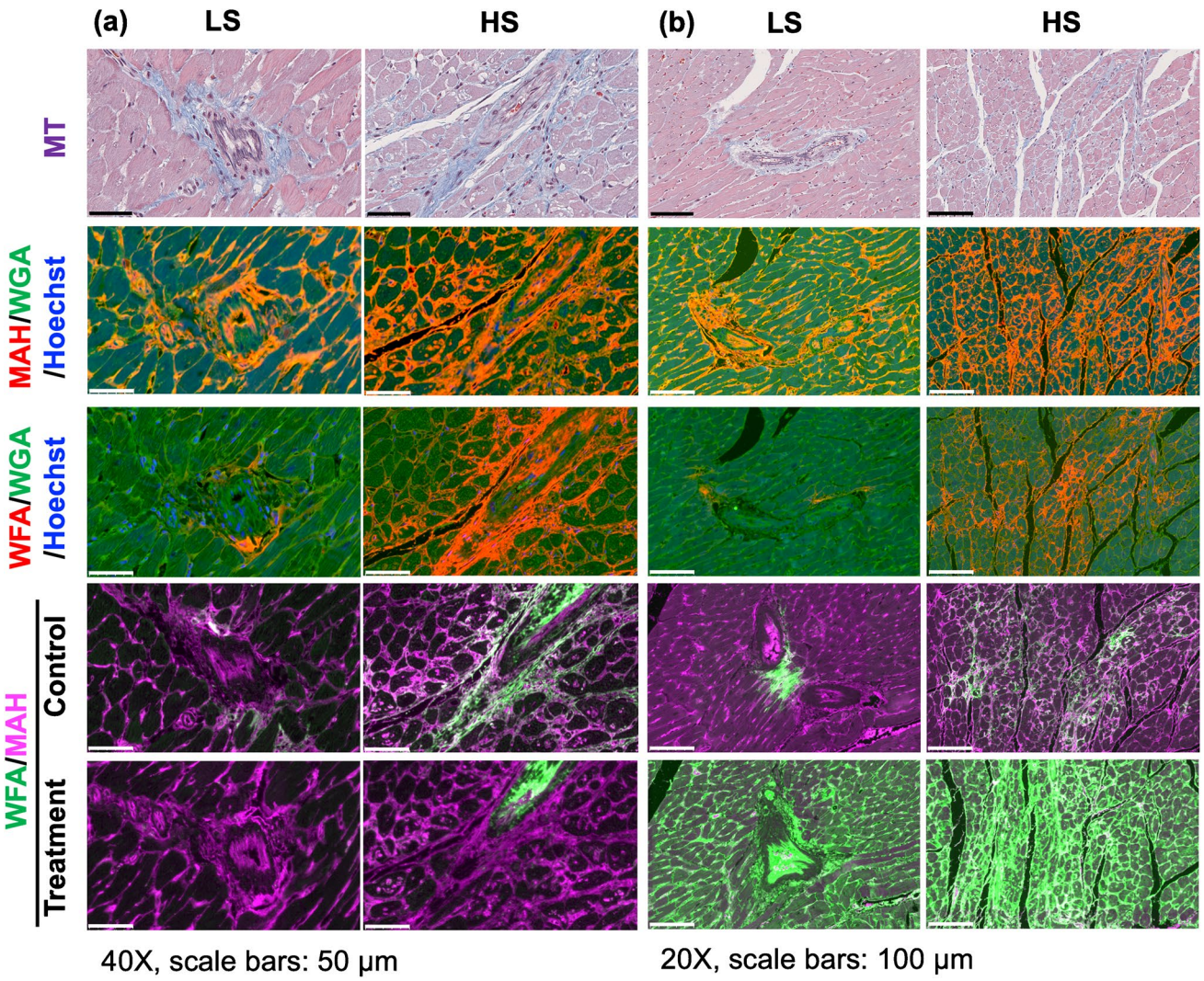


Fig. 3 Effects of glycosidase pretreatment on lectin histochemistry for *Wisteria floribunda* agglutinin (WFA) and *Maackia amurensis* hemagglutinin (MAH) in rat hearts. Serial tissue sections were either left untreated (control) or pretreated with PNGase F (**a**) or *O*-glycosidase with sialidase (**b**) to remove *N*-glycans and *O*-glycans, respectively. Representative images from perivascular fibrotic regions and interstitial regions of high-salt (HS) rat hearts and corresponding regions of low-salt (LS) rat hearts are shown. Masson's trichrome (MT) staining images of serial sections are also presented to indicate fibrotic regions. Wheat germ agglutinin (WGA) and Hoechst were used to visualize membranes and nuclei, respectively. Scale bars are as indicated

previously optimized for visualizing glomerular basement membranes. A brief explanation of the imaging contrast mechanism is provided here to aid interpretation of the LVSEM images. All three methods enable the clear identification of ECM components, such as collagen and reticular fibers, in LVSEM images (Supplementary Fig. 4). However, each staining method exhibited distinct advantages and limitations in the context of LVSEM imaging. Masson's PTA provided robust contrast of collagen-rich interstitial regions with relatively uniform background and is widely applicable in routine pathological assessment. Reticulin staining allowed visualization of fine reticular fibers but showed relatively higher background variability under LVSEM conditions. The modified PAM protocol generated strong basement membrane–associated contrast due to silver impregnation but required careful reaction control to avoid over-deposition of metal particles. Owing to the widespread use of MT to assess myocardial fibrosis in clinical pathology, Masson's PTA staining was used for subsequent analyses.

While the classical fluorescence microscopy shows largely distinct distribution with partial overlapping WFA and MAH staining (Fig. 4a), the high-resolution LVSEM-based ultrastructural imaging enabled spatial mapping of WFA and MAH signals within the fibrotic areas (Fig. 4). Notably, WFA epitopes were mainly observed in the interstitial regions, whereas MAH staining was observed

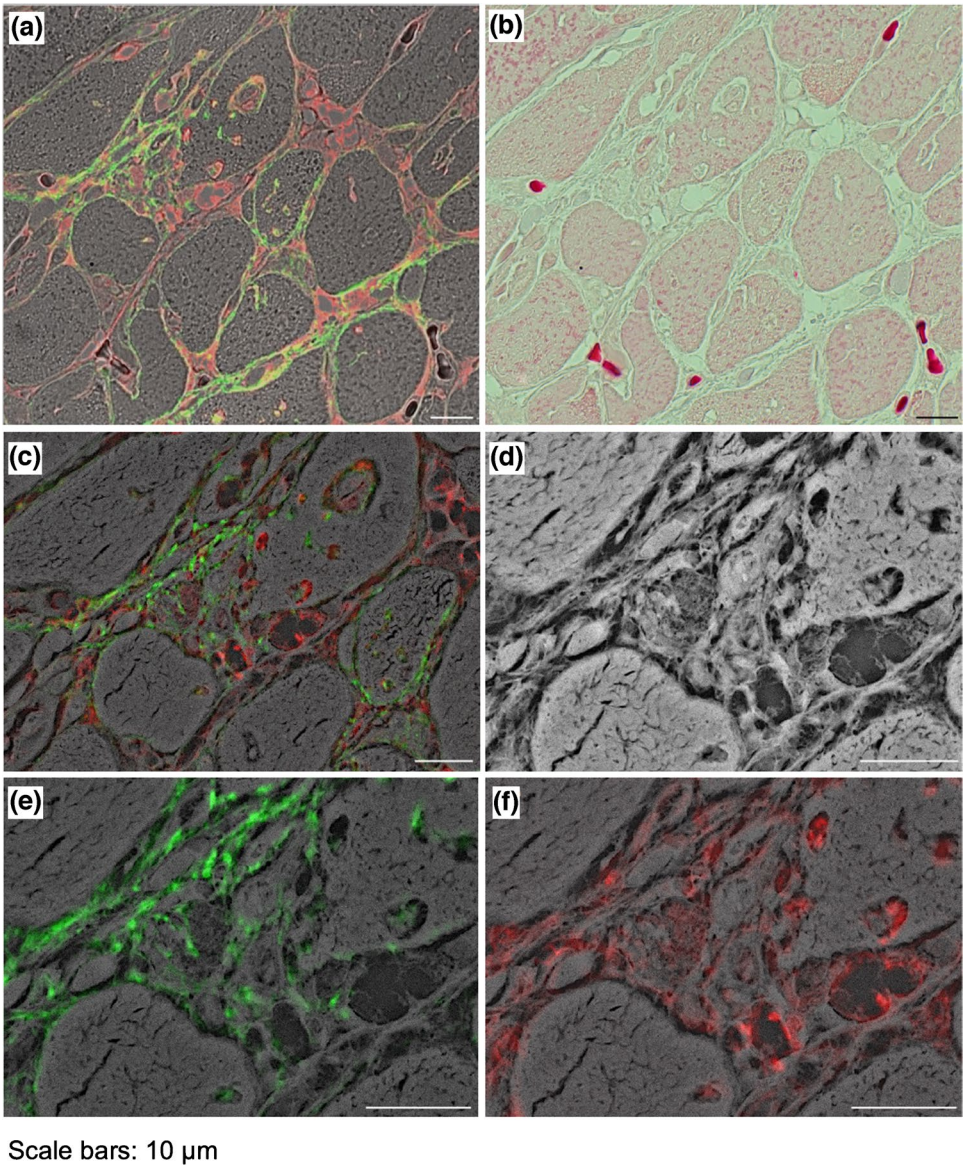


Fig. 4 Single-slide ultrastructural imaging of fluorescent lectin staining and low-vacuum scanning electron microscopy (LVSEM). **a** Merged image of fluorescent and bright field for *Wisteria floribunda* agglutinin (WFA; green) and *Maackia amurensis* hemagglutinin (MAH; red) staining. **b** Bright-field image of Masson's phosphotungstic acid (PTA) staining. **c** Merged image of fluorescence and LVSEM (magnified view of (**a**)). **d** LVSEM image corresponding to (**b**) (magnified view). **e** Merged image of WFA fluorescence and LVSEM (magnified view of (**c**)). **f** Merged image of MAH fluorescence and LVSEM (magnified view of (**c**))

in both the cardiomyocytes and extracellular regions. In LVSEM imaging, contrast enhancement of lectin labeling is achieved through heavy metal sensitization involving silver deposition and subsequent gold substitution. This process increases local electron density in a manner associated with the stained regions, thereby enhancing backscattered electron signals under low-vacuum conditions. Thus, image contrast arises from metal deposition rather than intrinsic fluorescence signals. Importantly, the present observations represent morphology-correlated visualization of lectin-reactive regions and should not be interpreted as a quantitative measurement of glycan abundance. Collectively, this imaging method demonstrates that LVSEM can visualize structurally contextualized patterns of WFA-reactive asialo-*N*-glycans and MAH-reactive sialo-*O*-glycans within fibrotic cardiac tissues. The method emphasizes spatial correlation between glycan-reactive regions and extracellular matrix architecture, providing a structural context for interpreting lectin signals detected by tissue glycome mapping. Thus, LVSEM-assisted lectin imaging should be interpreted primarily as a structurally contextualized visualization platform for glycan localization, while further biological validation of glycan functions will be required in future studies.

To date, dual-modality platforms for spatial *N*-glycomics analysis combining matrix-assisted laser desorption/ionization (MALDI) mass spectrometry imaging (MSI) with multiplexed antibody-based imaging have been reported [23]. However, the results are typically presented as two-dimensional heat maps of enzymatically released glycans and are generally limited to *N*-glycans. In addition, the accompanying fluorescence images often provide limited structural context for delineating fine tissue architecture, cellular

boundaries, and precise target localization. In contrast, the present results demonstrate the utility of tissue glycome mapping combined with LVSEM. This approach enables spatially detailed visualization of lectin-detected *N*- and *O*-glycan distribution within preserved tissue architecture, particularly in ECM-rich areas. This capability is particularly valuable for studying glycosylation changes associated with ECM remodeling during tissue fibrosis.

Evaluation of candidate glycoproteins carrying fibrosis-related glycans

The differential spatial patterns of WFA- and MAH-reactive glycans likely reflect differences in the expression of their carrier glycoproteins. To assess whether candidate glycoproteins could account for the spatial distribution of WFA- or MAH-reactive glycans, we performed double staining with these lectins.

We found that POSTN co-distributed with WFA in fibrotic areas (Fig. 5a), consistent with its upregulation in Dahl HS rat hearts (Supplementary Fig. 5) [13] and DCM mouse hearts [7]. Fibrillar collagen VI is one of eight collagens involved in fibrotic scar formation in human and experimental models [24]. COL6A6 was detected in the myocardial layer near the endocardial surface and in interstitial fibrotic areas, but it was mostly not co-distributed with WFA staining. In contrast, COL6A6 expressed in perivascular and microscar regions was randomly co-distributed with WFA (appearing as white color in Fig. 5b), implying that these proteins and glycans may vary depending on their expression patterns. These results suggest that POSTN and COL6A6 represent examples of candidate glycoproteins associated with fibrotic regions in this hypertensive heart failure model.

For double staining with MAH, candidate glycoproteins were selected from RNA-seq data available at [13] according to the criteria described in the "Materials and methods" section. Based on these criteria, CILP, THBS4, and NPPA were selected for validation (21 *N*-glycans at 5 sites and 12 *O*-glycans at 9 sites for CILP [25, 26], 6 *O*-glycans at 5 sites for THBS4 [26, 27], and 1 *O*-glycan at 1 site for NPPA [26]). Immunostaining with an anti-NT-proANP antibody shows signals primarily in cardiomyocytes, with little to no staining in extracellular regions, and does not colocalize with MAH staining (Supplementary Fig. 6). In contrast, CILP is observed in the extracellular region and is co-distributed with MAH, accumulating around regions with dense MAH staining corresponding to fibrotic regions (Fig. 6). A similar pattern is observed for THBS4. In LS hearts, CILP staining was nearly absent, and THBS4 expression was limited to minor MAH^+ regions. Previous studies have reported increased CILP expression and accumulation in fibrotic hearts [28]. Because CILP carries both *N*- and *O*-glycans, and spatial association observed by microscopy does not demonstrate a covalent linkage, further validation by mass spectrometry or antibody-overlay LMA is required to determine its contribution to MAH reactivity. THBS4 is known

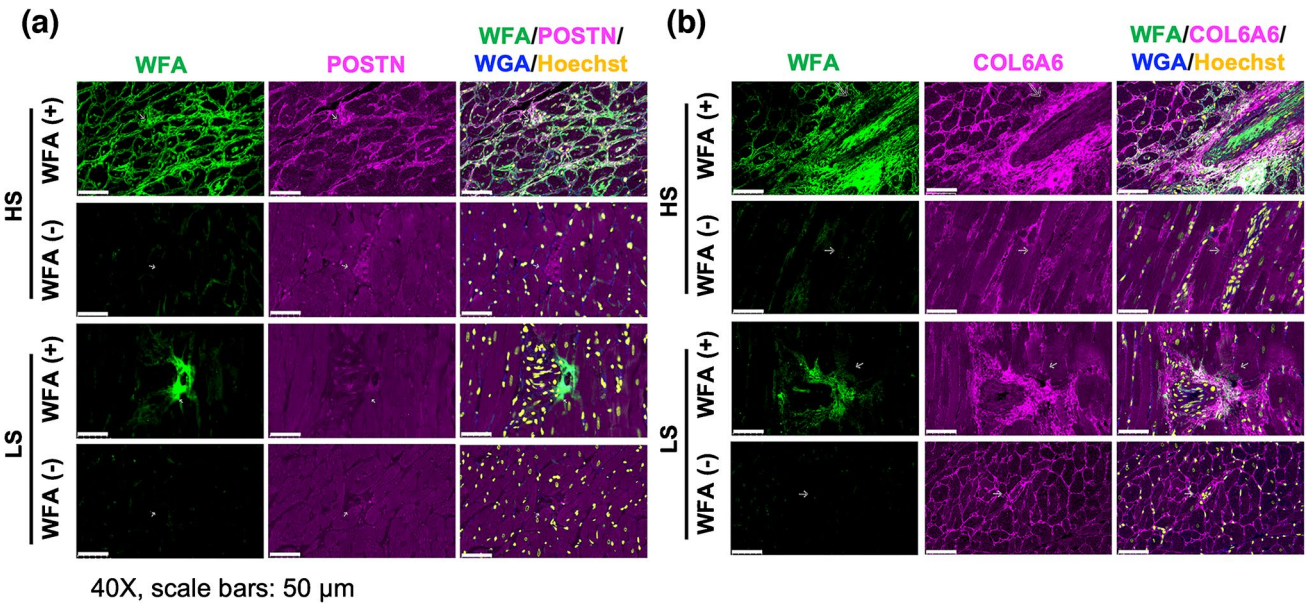


Fig. 5 Co-staining of *Wisteria floribunda* agglutinin (WFA) with candidate glycoproteins, POSTN (**a**) and COL6A6 (**b**) in rat hearts. Representative images from WFA-positive (WFA(+)) and WFA-negative (WFA(−)) regions of high-salt (HS) and low-salt (LS) rat hearts are shown. WFA staining (green) was compared with immunofluorescence staining of POSTN or COL6A6 (magenta), together with wheat germ agglutinin (WGA) and Hoechst staining. WGA and Hoechst were used to visualize membranes and nuclei, respectively. Scale bars are as indicated

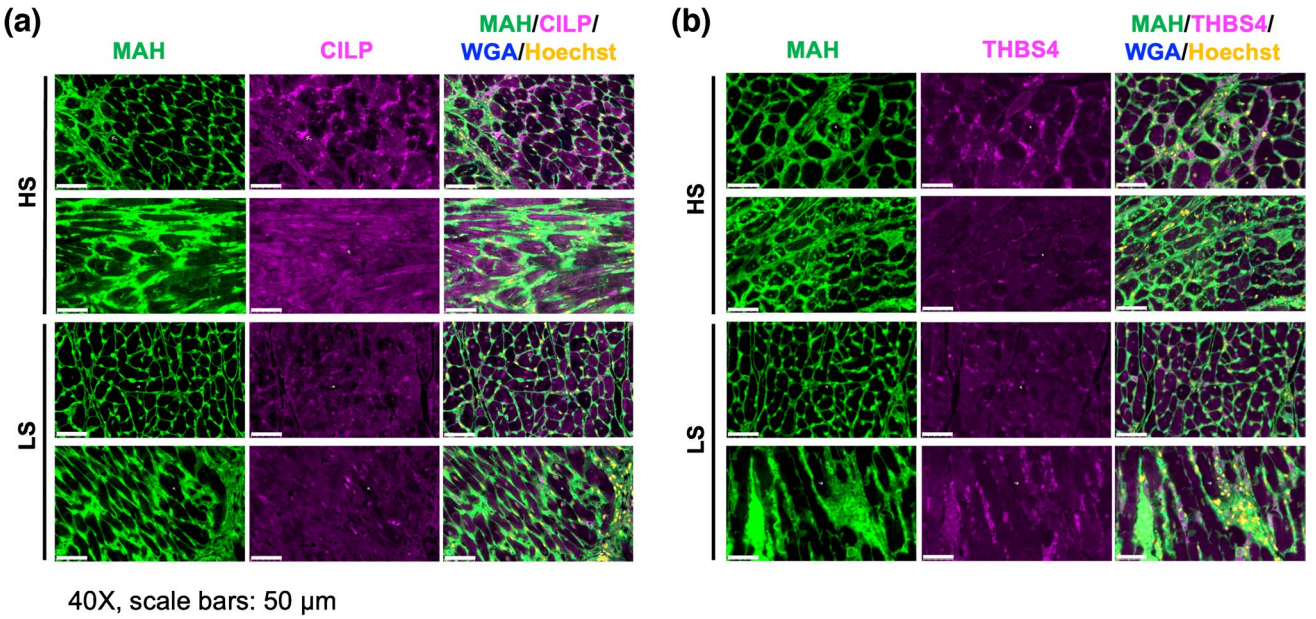


Fig. 6 Co-staining of *Maackia amurensis* hemagglutinin (MAH) with candidate glycoproteins, cartilage intermediate layer protein 1 (CILP) (**a**) and thrombospondin 4 (THBS4) (**b**) in rat hearts. Representative images from non-fibrotic and fibrotic regions of high-salt (HS) and low-salt (LS) rat hearts are shown (upper rows, HS; lower rows, LS). MAH staining (green) is compared with immunofluorescence staining of CILP or THBS4 (magenta), together with wheat germ agglutinin (WGA) and Hoechst staining. WGA and Hoechst were used to visualize membranes and nuclei, respectively. Scale bars are as indicated

to participate in myocardial remodeling under pressure overload [28–31], supporting its selection as a fibrosis-associated glycoprotein in this model. Similar to CILP, further validation of THBS4 glycan structures is required. Given the broad occurrence of sialylated *O*-glycans, it is likely that multiple glycoproteins contribute to the observed MAH-reactive signals. NT-pro-ANP did not overlap with MAH staining, consistent with its storage within cardiomyocytes as a precursor protein [32, 33]. In rats, four *O*-glycosylation sites in NT-proANP and two in mature ANP have been reported [32]. In this study, fluorescence staining revealed NT-proANP localization within cardiomyocytes, distinct from interstitial MAH^+ regions, suggesting that NT-proANP is not the main contributor to MAH staining. Nevertheless, LVSEM imaging reveals MAH positivity in granular structures within cardiomyocytes (Fig. 4), possibly corresponding to secretory vesicles containing pro-ANP with *O*-glycosylation sites recognized by MAH. Further confirmation is required to verify this association.

Taken together, the lectin signals observed in WFA^+ and MAH^+ fibrotic regions may reflect a combination of increased abundance of ECM glycoproteins and alterations in their glycosylation patterns. Lectin–antibody double staining is useful for visualizing the spatial distribution of specific glycans and proteins within tissue sections. However, this approach cannot identify the specific carrier proteins or determine their precise glycan structures. Importantly, the microscopic co-localization of lectin signals with ECM glycoproteins does not indicate direct molecular linkage between the detected glycans and the co-localized proteins. Therefore, additional analyses such as lectin affinity pull-down assays or glycoproteomic approaches are required to identify glycan-carrying proteins and characterize their glycan structures. Furthermore, lectin recognition reflects the glycan signature patterns rather than precise glycan structures or site-specific modifications. Accordingly, integration with mass spectrometry–based glycomics and glycoproteomics will be important for future studies to achieve molecular-level characterization of fibrosis-associated glycans. In addition, the present workflow is particularly suited for ECM-rich tissues where conventional nucleic acid–based spatial omics approaches are less informative, highlighting its potential utility for spatial glycomic analysis of fibrotic microenvironments.

Conclusion

This study establishes a hierarchical lectin-based spatial glycomics workflow integrating glycomic signal detection, spatial localization, and ultrastructural validation. By combining LMD-assisted lectin microarray–based tissue glycome mapping, lectin histochemistry, and single-slide LVSEM imaging, this workflow enables spatial mapping of glycan signals within fibrotic ECM architecture. Using this analytical workflow, we observed distinct spatial distributions of WFA-reactive asialo-*N*-glycans and MAH-reactive sialo-*O*-glycans in fibrotic cardiac tissues,

and demonstrated that WFA and MAH serve as effective histochemical probes for detecting these glycan patterns. Beyond these specific biological observations, this workflow provides an analytical framework for investigating spatial glycan remodeling in ECM-rich tissues where conventional nucleic acid–based spatial omics approaches have limited applicability, thereby facilitating broader spatial glycomics studies in fibrotic diseases.

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Author contribution Conceptualization, C.N.-O. and A.K.; formal analysis, P.B., K.K., M.B.; resources, T.N. and T.I.; writing—original draft preparation, P.B.; writing—review and editing, K.A.; supervision, C.N.-O. and A.K.; funding acquisition, P.B., C.N.-O., K.K. All authors have read and agreed to the published version of the manuscript.

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Data availability All data supporting the findings of this study are provided in the Supplementary Materials.

Declarations

Ethics approval, source of biological material, and statement on animal welfare All applicable international and institutional guidelines for animal care and use were followed strictly. All animals were treated according to an experimental protocol approved by the Institutional Animal Care and Use Committee of the University of Tsukuba, in compliance with the Guide for the Care and Use of Laboratory Animals (reference number: 20-146). All biological samples were obtained from Dahl-Iwai S rats (Japan SLC, Shizuoka, Japan), maintained and provided by the University of Tsukuba as part of previously approved animal studies [13].

Conflict of interest The authors declare no conflict of interest.

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